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Week 7 Prqactical

VM Insights and enhanced monitoring were enabled on Net731-VM using the Azure Monitor agent. Collected data including CPU utilization, disk latency, and availability metrics is stored in the Log Analytics Workspace (net731-workspace) in South Africa North. Enabling monitoring has minimal cost implications on a free tier subscription but may incur charges in production based on data ingestion volume.

The Percentage CPU metric showed an average of 7.6% over the last 24 hours, which is expected for an idle VM with no active workloads running. A spike was visible during the initial deployment and configuration period, which is normal behaviour.

Network In Total and Network Out Total metrics were unavailable for this VM type (Standard_B2as_v2) in the selected region. OS Disk Latency was used as an alternative I/O metric. In a production environment with a supported VM type, Network In/Out metrics would be available through the Virtual Machine standard namespace.

The OS Disk Latency metric showed 0.79ms average, indicating very low disk activity consistent with an idle VM. Data Disk Read Bytes/Sec was also monitored. High disk write activity would be expected during software installations, OS updates, or database transactions.

A CPU alert rule was configured to trigger when average CPU usage exceeds 80% over a 5-minute period. The 80% threshold was chosen because sustained CPU usage above this level typically indicates a performance bottleneck or runaway process that requires investigation. The alert notifies via email through the TechBridge-AlertGroup action group, ensuring rapid response before the issue impacts users.

Network In Total and Network Out Total metrics were unavailable for this VM type (Standard_B2as_v2) in the selected region. As an alternative, OS Disk Latency and Data Disk Read Bytes/Sec were used to demonstrate I/O monitoring capability. In a production environment with a supported VM type, Network In/Out metrics would be available through the Virtual Machine standard namespace.

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Available

No outages

2 events

Run all tests

Checks for common network security rules at RDP, HTTP, HTTPS.

For more network diagnostic tools, visit Network Watcher

Unlock enhanced monitoring

Upgrade to OpenTelemetry metrics for more accurate, detailed, and actionable insights into the performance of your VMs and applications. You'll still have access to basic host-level metrics even if you don't upgrade.

ConfigureRemind me laterLearn more

Metrics

CPU / Availability / Memory

Availability

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Diagnostic setting

SaveDiscardDeleteFeedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

Diagnostic setting nameNet731-NSG-Diagnostics

Logs

Category groups

allLogs

Categories

VM protection alerts

Metrics

AllMetrics

Destination details

Send to Log Analytics workspace

Subscription

Azure subscription 1

Log Analytics workspace

net731-workspace (southafricanorth)

Archive to a storage account

Stream to an event hub

Send to partner solution

Diagnostic setting ...

Updating diagnosti
Successfully updated di

Save Discard Delete Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

Diagnostic setting name Net731-VNet-Diagnostics

Logs

Category groups

☐ allLogs

Categories

☐ VM protection alerts

Metrics

☒ AllMetrics

Destination details

☒ Send to Log Analytics workspace

Subscription

Azure subscription 1

Log Analytics workspace

net731-workspace (southafricanorth)

☐ Archive to a storage account

☐ Stream to an event hub

☐ Send to partner solution

Monitor | Metrics ...

How can I programmatically access Azure metrics?

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New chart Refresh Share

Local Time: Last 24 hours (Automatic - 5 minutes)

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Avg Percentage CPU for Azure subscription 1 in South Africa North region by ResourceId where ResourceGroupName = 'Net731'

Add metric Add filter Apply splitting

Line chart Drill into Logs New alert rule Save to dashboard

Scope Azure subscription 1

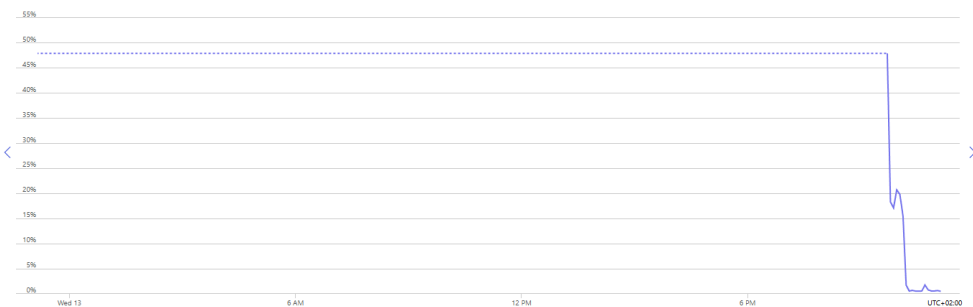
Metric Namespace Virtual machine stand...

Metric Percentage CPU

Aggregation Avg

ResourceGroup... = Net...

Split by = ResourceId



Net731-VM | 8.0087%

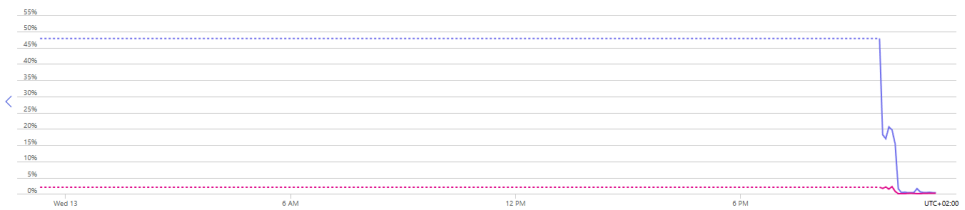
Avg Percentage CPU and Avg OS Disk Latency (Preview) for Azure subscription 1 in South Africa North region by ResourceId where ResourceGroupName = 'Net731'

+ Add metric Add filter Apply splitting

Line chart Drill into Logs New alert rule Save to dashboard

Scope	Metric Namespace	Metric	Aggregation
Azure subscription 1	Virtual machine stand...	Percentage CPU	Avg
Azure subscription 1	Virtual machine stand...	OS Disk Latency (Prev...	Avg

ResourceGroup... = Net... Split by = ResourceId

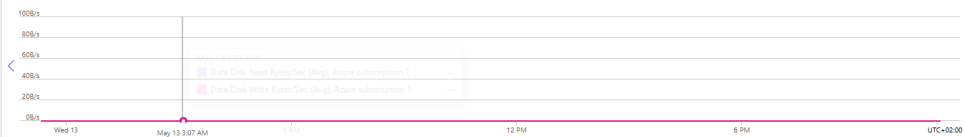


Avg Data Disk Read Bytes/Sec and Avg Data Disk Write Bytes/Sec for Azure subscription 1 in South Africa North region by ResourceId where ResourceGroupName = 'Net731'

+ Add metric Add filter Apply splitting

Line chart Drill into Logs New alert rule Save to dashboard

Azure subscription 1, Data Disk Read Bytes/Sec, Avg Azure subscription 1, Data Disk Write Bytes/Sec, Avg ResourceGroup... = Net... Split by = ResourceId



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+ Create Columns

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Name

Net731-CPU-Alert

Net731-CPU-Alert

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Location (...)

Subscription (...)

Subscription ID

Tags (...)

: Net731

: Global

: Azure subscription 1

: 62c914fe-14ed-4504-a597-7a1aad22aa17

: Add tags

Severity

Description

: 2 - Warning

: Alert when CPU exceeds 80% for 5 minutes

Scope

Resource

Hierarchy

net731-vm

Azure subs...

net731

Actions

Name

Contains actions

Net731-AlertGroup

1 Email

Conditions

Name

Time series monitored

Estimated monthly cost

Percentage CPU

1

\$0.10

